

Git Training Manual for Attendance Management System

1. Git Workflow Overview

Branch Structure:

main → production-ready code

dev → integration branch

feature/* → every task gets its own branch

2. Clone Repository

```
git clone https://github.com/your-team/attendance-system.git
```

```
cd attendance-system
```

3. Create Feature Branch

```
git checkout dev
```

```
git pull origin dev
```

```
git checkout -b feature/auth-module
```

4. Add, Commit, Push

```
git add .
```

```
git commit -m "Added JWT login API"
```

```
git push -u origin feature/auth-module
```

5. Pull Request Workflow

Create PR from feature/auth-module → dev.

6. Merge vs Rebase

Merge:

```
git checkout dev
```

```
git merge feature/auth-module
```

Rebase:

```
git checkout feature/auth-module
```

```
git rebase dev
```

7. Reset Commands

Soft reset:

```
git reset --soft HEAD~1
```

Mixed reset:

```
git reset HEAD~1
```

Hard reset:

```
git reset --hard HEAD~1
```

8. Log Commands

```
git log --oneline
```

```
git log --oneline --graph --all
```

```
git diff
```

```
git blame filename.js
```

9. Real Project Example

Feature: Attendance Create API

```
git checkout dev
```

```
git checkout -b feature/attendance-create-api
```

Coding...

```
git add .
```

```
git commit -m "Implemented attendance create API"
```

```
git push -u origin feature/attendance-create-api
```

Open PR on GitHub

10. Daily Developer Workflow

```
git checkout dev
```

```
git pull origin dev
```

```
git checkout feature/auth-module
```

```
git rebase dev
```

11. Branch Cleanup

```
git branch -d feature/auth-module
```

```
git push origin --delete feature/auth-module
```